

***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2018***

**CE/EE/ME/SE GE 15-1103 A & CS/EC/IT GE 15-1203 B ENGINEERING MECHANICS
(2015 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

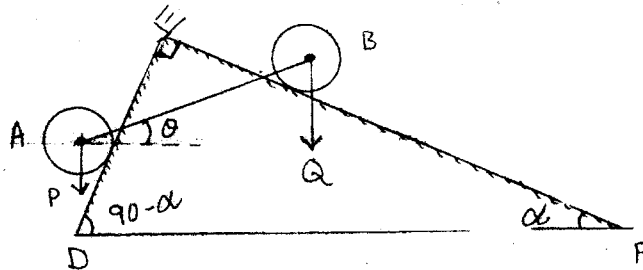
- I. (a) State and explain the 'Theorems of Pappus'.
- (b) State the laws of friction.
- (c) Two forces P and Q act at right angles and the magnitude of the resultant is $\sqrt{25}$ N. If they act at 60° to each other, the magnitude of their resultant is $\sqrt{33}$ N. Find the magnitude of forces P and Q.
- (d) State and prove parallel axis theorem in Moment of Inertia.
- (e) What is 'Redundancy' in analysis of truss?
- (f) Derive expressions for velocity and acceleration of a particle executing simple harmonic motion.
- (g) Explain how momentum gets conserved in collision of two bodies.
- (h) Derive an expression for the trajectory of a projectile, projected from a horizontal plane with initial velocity 'u' and angle of projection ' α '.
- (i) State and prove the work-energy principle in curvilinear translation.
- (j) Explain D'Alemberts principle in rotation.



PART B

(4 × 10 = 40)
(10)

II.



Two rollers of weight 'P' and 'Q' are connected by a flexible string 'AB'. The rollers rest on two mutually perpendicular planes DE and EF as shown in figure. Find the tension in the string and angle ' θ ' that it makes with the horizontal, when the system is in equilibrium. Given $P = 50$ N, $Q = 100$ N and $\alpha = 30^\circ$.

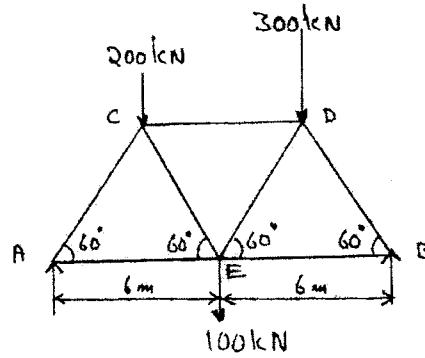
OR

- III. A circular hole is cut at the corner of a square whose diameter is equal to half the side of a square. Find the centroid of the remaining portion. (10)

(P.T.O.)

IV. (a)

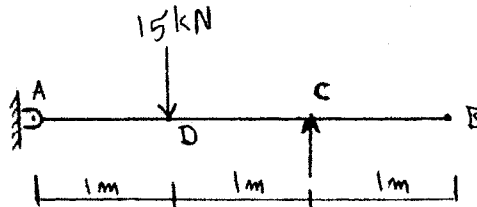
(5)



A loaded truss is shown in figure. Determine the forces in the members of the truss.

(b)

(5)



A beam AB, 3 m long is hinged at 'A' and supported at 'C'. Determine the reaction at 'C' when it is loaded 15 kN at 'D' by using principle of virtual work.

OR

V. (a) Differentiate between polar moment of inertia and product of inertia. (5)

(b) A stone of 270 N is to be carried by two persons, one stronger than the other. The weight is tied to a uniform rod of negligible weight having 6 m length. They are planning to lift the rod holding at its extreme ends. The stronger man can carry 180 N. Find the position on the rod where the stone is to be tied so that they can lift the rod along with the stone. (5)

VI. (a) Equation of motion of a moving body along a straight line is given by $S = 4t^3 - 8t + 20$, where 'S' is in metre and 't' in seconds. Find: (5)

- Time taken by the body to reach a velocity of 100 m/s.
- Acceleration of the body, when velocity is 50 m/s.
- Distance travelled by the body in 4 seconds.

(b) Determine the velocity and acceleration of a piston of a heat engine when it is at a distance of 75 cm from the centre and crank rotates at 120 rpm. Take stroke length as 200 cm. (5)

OR

VII. (a) A ball of mass 'm' collides with another identical mass which is at rest and first ball is brought to rest. If the loss in kinetic energy due to impact is $\frac{1}{3}$ (5)

of the initial kinetic energy of the first mass, find the value of 'e'.

(b) A bullet travelling horizontally with a velocity of 700 m/s and weighing 0.25 N strikes a wooden block of weight 50 N resting on a rough horizontal plane. The coefficient of friction between the floor and the block, $\mu = 0.5$. Find the distance through which the block is displaced from the initial position. (5)

VIII. A constant torque of 2500 Nm is exerted on a crank shaft to start the engine. If there is a uniform resisting torque of 1500 Nm, find the speed attained by the engine after one minute from the start. Take mass, $m = 1760$ kg and radius of gyration, $k = 1$ m for the flywheel mounted on the engine shaft. (10)

OR

IX. A right circular cylinder of mass 'm' and radius 'r' is suspended from a cord that is wound around its circumference. If the cylinder is allowed to fall freely, find the acceleration of its mass centre and tension in the cord. (10)